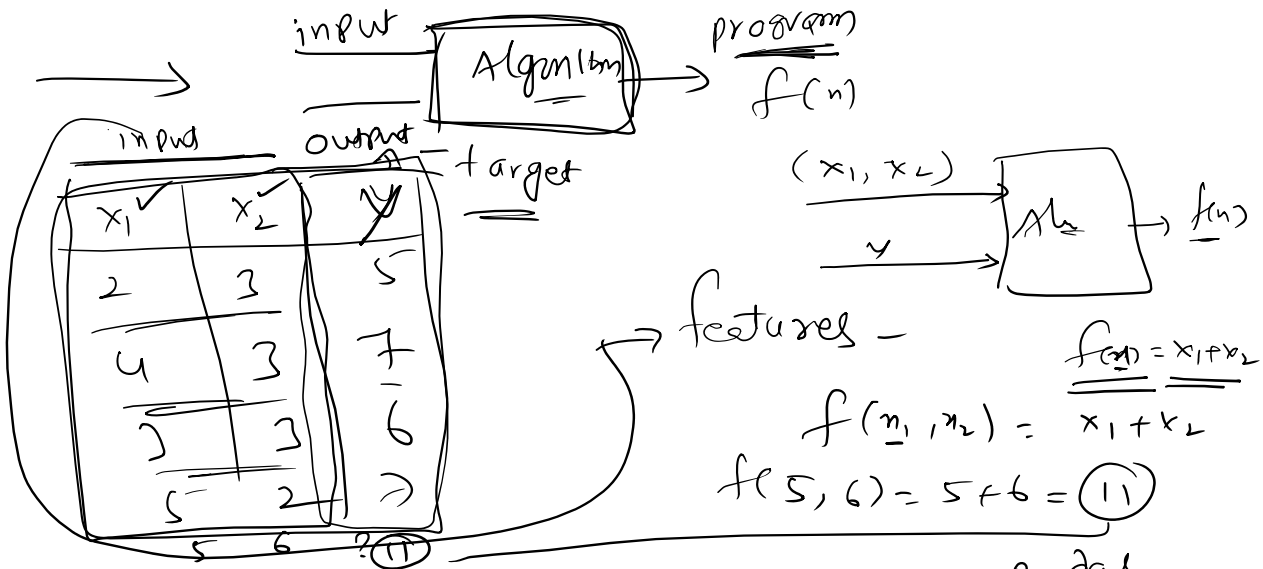
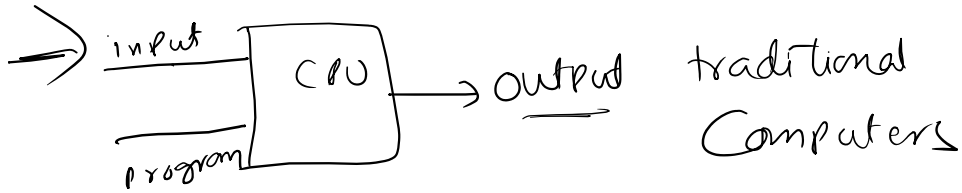
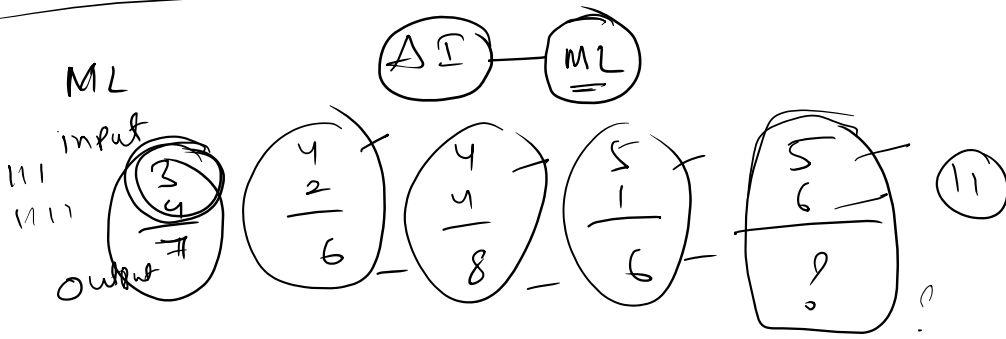


Introduction to Classical Machine Learning:



Sklearn → scikit - learn

Algorithms

pandas
numpy

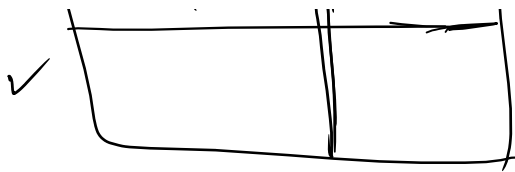
① Linear Regression

① ML → Supervised Learning ✓

→ Unsupervised Learning ✓

→ Reinforcement Learning ✓





classification

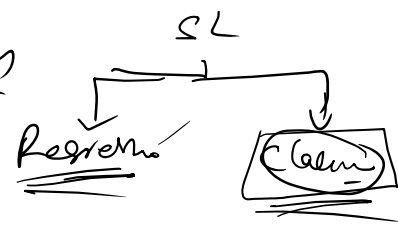
x min	y max
20	21
22	25
32	26
30	35

million

Are	Q	E	Salary
			1,000.0
			2,000
			1500
			1

Are	Salary	Delta	gm
			yes
			no
			no
			we

classification



SL SW PL PW SM

miscellaneous

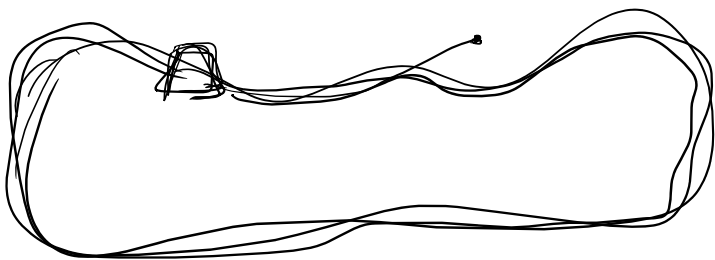
Unsupervised learning

moviename	Genre =
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features

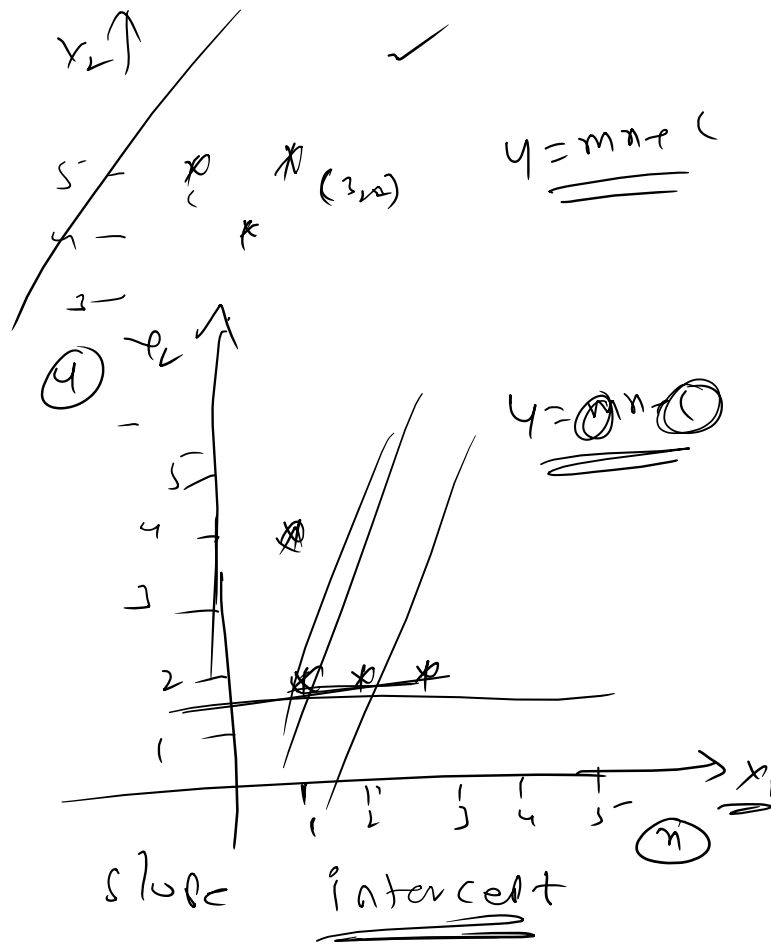
target

✓ Reinforcement learning → Award & punishment

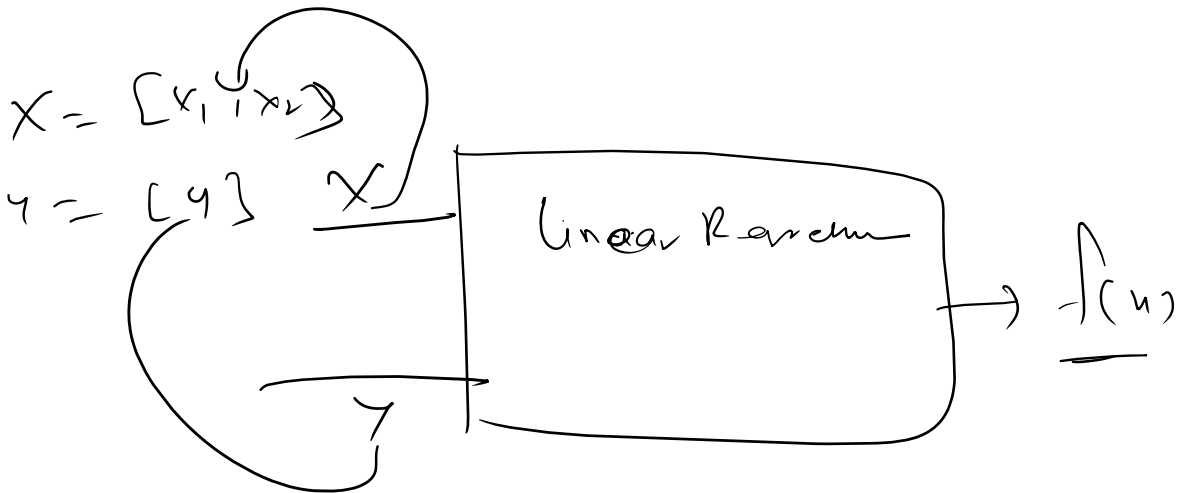


1) Linear Regression

x_1	x_2	y
1	2	2
2	2	5
1	5	5
2	2	4



Linear Regression



$y \rightarrow y = mx + c$

$y \rightarrow y = m_1 x_1 + m_2 x_2 + c$

$x_1, x_2, x_3, y \rightarrow y = m_1 x_1 + m_2 x_2 + m_3 x_3 + c$

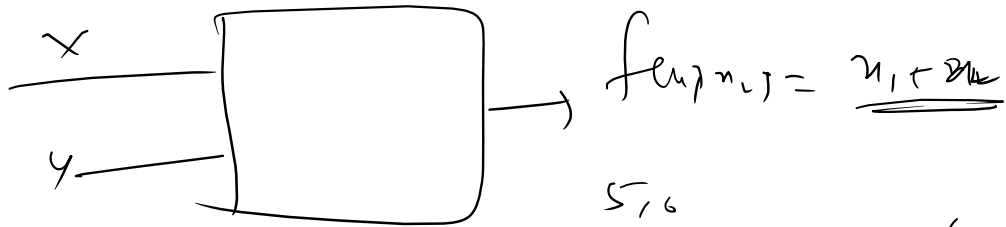
$y = \underset{\substack{\downarrow \\ \text{coef}_1}}{m_1} x_1 + \underset{\substack{\downarrow \\ \text{coef}_2}}{m_2} x_2 + \underset{\substack{\downarrow \\ \text{intercept}}}{c}$

$[1, 1] \text{ Coeff}$

$[0] \text{ intercept}$

$$y = \frac{1}{x_1 + x_2} + 1(x_1) + 0$$

$$y = x_1 + x_2$$



$$5, 6$$

$$f(5, 6) = 5 + 6 = 11$$

x_1	x_2	y target
1	2	3
1	4	5
1	5	6
2	2	4

build a algorithm

→ SL → Regression → Linear Regression

Sklearn → Linear-model

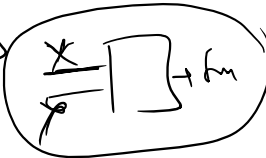
from sklearn.linear_model import LinearRegression

import pandas as pd

$X = [x_1, x_2]$

$lo = \text{LinearRegression}()$

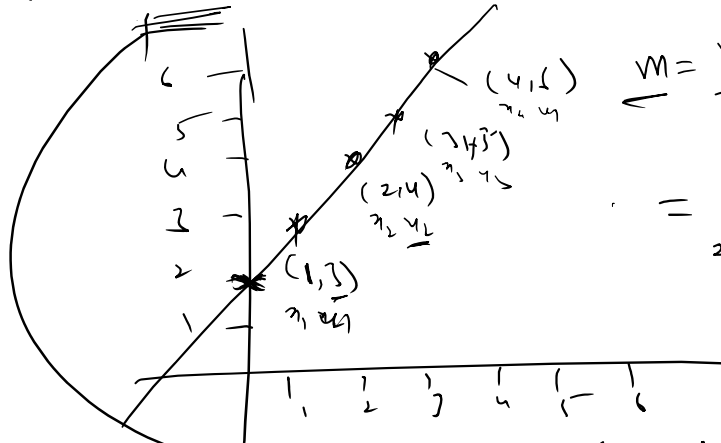
$y = [y]$



$lo.fit(X, y)$

x_1	y
1	3
2	4
3	5
4	6

$$f(x) = x + 2 \rightarrow y = mx + c$$



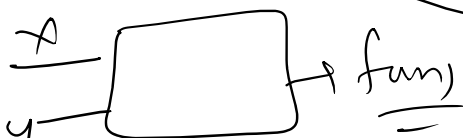
$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{4 - 3}{2 - 1} = 1$$

$$c = 2$$

$$y = 1(x) + 2$$

$$= x + 2$$



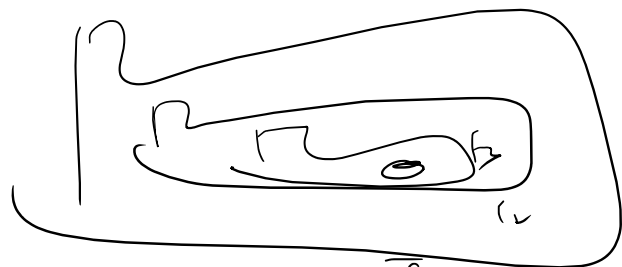
f → fun

- import a algorithm
- from sklearn.linear_model import Linear
- df
- x, y
- fit() → predict

ML → SL, UL, RF

Regression Classification

↓
(lr) →



from

f

from f1, f2, f3 import all

from sklearn.linear_model import lr